Tzu-Yao **Lin**

Joint PhD candidate

tzu-yao.lin@kuleuven.be | https://tzuyao-lin.github.io/ | tzuyao-lin | tzu-yao-nick-lin | 0000-0002-2261-448X

I am a quantitative psychologist and joint PhD candidate at KU Leuven and Maastricht University. My research focuses on reliability in multilevel and intensive longitudinal data, with broader interests in longitudinal and multilevel modeling and mathematical and computational models of judgment and decision-making.

Education

KU Leuven and Maastricht University — *Leuven, Belgium and Maastricht, the Netherlands*

Joint PhD candidate — *2022–Present*

- Research Group of Quantitative Psychology and Individual Differences; Department of Methodology and Statistics

National Taiwan University — *Taipei, Taiwan*

MS in Psychology — *September 2019–June 2022*

- Overall GPA: 4.02/4.30

National Taiwan University — *Taipei, Taiwan*

BS in Psychology, minor in Forestry — *September 2014–June 2019*

- Overall GPA: 4.12/4.30; minor in the School of Forestry and Resource Conservation

Publications

Lin, T.-Y., Tuerlinckx, F., & Vanbelle, S. — *Manuscript under review*

An intraclass correlation coefficient is a correlation: A tutorial — *Under review*

- Preprint

Lin, T.-Y., Tuerlinckx, F., & Vanbelle, S. — *Psychometrika*

Single-Item Reliability for Intensive Longitudinal Data: A Three-Level Linear Mixed-Effect Autoregressive Modeling Approach — *2026*

- DOI

Lin, T.-Y., & Hsu, Y.-F. — *Journal of Mathematical Psychology*

Incorporating the Luce–Krantz threshold model into cultural consensus theory for ordinal categorical data — *2026*

- DOI

Lo, C., Lin, T.-Y., Werneck, G. L., Domingues, C., Saavedra, R. da C., Moore, M., Souza, L. E. P. F., Regazzi, L., & Lomazzi, M. — *Journal of Global Health Economics and Policy, 5, e2025034*

The impact of the COVID-19 pandemic on vaccine sentiment among Brazilian healthcare workers: A cross-sectional online survey study — *2025*

- DOI

Rao, Y., Yu, E. Y.-W., Lin, T.-Y., Chen, Y., Qin, Y., Koster, A., Eussen, S. J. P. M., Bosma, H., Berendschot, T. J. M., van der Kallen, C., van Greevenbroek, M., de Galan, B., Zeegers, M., & Wesselius, A. — *European Journal of Nutrition, 64, 263*

Interaction between coffee consumption and polygenic risk score in relation to diabetes: Insights from The Maastricht Study — *2025*

- DOI

Lin, T.-Y., Tuerlinckx, F., & Vanbelle, S. — *Psychological Methods*

Reliability for multilevel data: A correlation approach — *2025*

- DOI

Yu, K., Lin, T.-Y., Zaman, J., Tuerlinckx, F., & Vanhasbroeck, N. — *Behavior Research Methods, 57(5), 127*

Consistency of perceptual response variability in size estimation and reproduction tasks — *2025*

- DOI

Conference Presentations

IMPS 2026 Annual Meeting — *Yonsei University, Seoul, South Korea*

Reliability for intensive longitudinal data: A three-level linear mixed-effect autoregressive modeling approach — *Jul 2026*

- Oral presentation

IOPS Winter Conference 2025 — *Utrecht University*

Reliability for intensive longitudinal data — *Dec 2025*

- Oral presentation

OKP Research Seminar, Research Group on Quantitative Psychology and Individual Differences — *KU Leuven*

Reliability for intensive longitudinal data — *Sep 2025*

- Oral presentation

IMPS 2024 Annual Meeting — *Prague University of Economics and Business, Prague, Czech Republic*

Reliability in intensive longitudinal studies: A state-space vs. linear mixed model approach — *Jul 2024*

- Oral presentation

OKP Research Seminar, Research Group on Quantitative Psychology and Individual Differences — *KU Leuven*

Reliability for multilevel data: A correlation approach — *Mar 2024*

- Oral presentation

Leuven Statistics Day: Collaboration — *Leuven, Belgium*

Reliability for multilevel data: A correlation approach — *Nov 2023*

- Oral presentation

MathPsych/ICCM/EMPG 2023 — *University of Amsterdam, Roeterseiland campus, Amsterdam, the Netherlands*

Incorporating the Luce–Krantz threshold model into the cultural consensus theory for ordinal categorical data: A simulation study — *Jul 2023*

- Oral presentation

EAM 2023 Congress of Methodology — *Ghent University, Campus Dunant, Ghent, Belgium*

Reliability for multilevel data: A correlation approach — *Jul 2023*

- Oral presentation

Colloquia Methodology and Statistics, Department of Methodology and Statistics — *Maastricht University*

Reliability for multilevel data: A correlation approach — *Jun 2023*

- Oral presentation

Awards

Dean's Award

College of Science, National Taiwan University — *2022 & 2019*

- The top three of the class graduating for outstanding Master's research at the academic year 2021-2022

Dean's Award

College of Science, National Taiwan University — *2022 & 2019*

- The top 10% of the class graduating for outstanding scholastic achievement at the academic year 2018-2019

Professor Fa-Yu Cheng Memorial Scholarship

Department of Psychology, National Taiwan University — *2019*

- The top prize of undergraduate psychology students for outstanding academic performance at the academic year 2018-2019

Academic Excellence Award

Department of Psychology, National Taiwan University — *Spring 2017*

- Awarded to students ranking top 5% in the department in the semester

Department of Forest and Resources Conservation, National Taiwan University — *Spring 2015*

- Awarded to students ranking top 5% in the department in the semester

Research Experience

Neural correlates of emotion as an arbitrator to reconcile the conflicts within dual-process in the context of decision making under risk — *Prof. Yung-Fong Hsu*

Research Assistant — *March 2021–Present*

- We studied the mechanism of dual-processing (heuristic and analytical) conflicts and the role of emotion when making a risky decision. I helped design the experiment of creating a 'Mixed Prospect Binary Lottery' to elicit the subject's utility functions and estimate the level of loss aversion.

Neural correlates of emotion as an arbitrator to reconcile the conflicts within dual-process in the context of decision making under risk — *Prof. Yung-Fong Hsu*

Research Assistant — *March 2021–Present*

- We will utilize PET-fMRI and MRS-fMRI, the state-of-the-art neuroimaging techniques, to examine the effects of neurotransmitter and BOLD signals in a risky situation.

The cultural consensus theory extends with the threshold theory: A study of ordinal categorical data analysis — *Prof. Yung-Fong Hsu*

Research Assistant — *August 2019–July 2020*

- We incorporated the Luce-Krantz threshold theory into cultural consensus theory to deal with ordinal categorical data. I axiomatized our new model and used the hierarchical Bayesian framwork for the inference.

From mind reading to mind sharing: A study on neural correlates of cognitive and affective Theory of Mind and their applications to salesforce enhancement — *Prof. Heng-Chiang Huang*

Research Assistant — *January 2018–December 2019*

- We want to know if a seller has greater abilities of cognitive and affective Theory of Mind will they have a better selling performance. We designed two different games to test those abilities of Theory of Mind with fMRI and MEG.

From mind reading to mind sharing: A study on neural correlates of cognitive and affective Theory of Mind and their applications to salesforce enhancement — *Prof. Heng-Chiang Huang*

Research Assistant — *January 2018–December 2019*

- I have played a critical role in this project by taking the initiative to learn Brainstorm (a MATLAB toolbox for M/EEG data analysis) and analyzing event-relative field and source estimation for our data. I also used the machine learning approach to classify the neural representation of numerosity (the set size of a group of items).

Coalition without trust: The intra-brain connectivity and inter-brain synchronization of herd behaviors in an economic bubble game — *Prof. Yu-Ping Chen*

Research Assistant — *January 2018–Present*

- We observed pairs of subjects' stock trading behaviors in a simulated economic bubble game. The breakthrough in this project was that we connected two MRI scan machines and simultaneously collected both subjects' BOLD signals.

Coalition without trust: The intra-brain connectivity and inter-brain synchronization of herd behaviors in an economic bubble game — *Prof. Yu-Ping Chen*

Research Assistant — *January 2018–Present*

- I learned the fMRI analysis, established the analytical procedure, and recorded tutoring videos for our research team.

Research of positive and negative memorable tourism experience — *Prof. Chia-Pin Yu*

Learning Assistant — *September 2016–January 2017*

- Assisted in data collection and qualitative interviews about the memorable tourism experience.

Characteristics of the evapotranspiration of a Japanese cedar forest in Xitou in Taiwan — *Researcher Sophie Laplace*

Research Assistant — *February 2015–January 2016*

- Assisted in collecting the evapotranspiration data of tree sap flow in the Xitou experimental forest.

Teaching Assistant Experience

Fall 2021	**PSY7001: Experimental Design**
Fall 2020	**PSY7001: Experimental Design**
Spring 2020	**PSY1004: Statistics in Psychology and Education II**, 4.33/5.00
Fall 2019	**PSY1003: Statistics in Psychology and Education I**, 4.40/5.00

Graduate-level Statistics and Modeling Coursework

Fall 2021	**CSIE5043: Machine Learning**, A
Spring 2021	**MATH7610: Multivariate Statistical Analysis**, A+
Fall 2020	**PSY7277: Neural and Behavioral Modeling**, A+
Fall 2020	**MATH7606: Regression Analysis**, A
Spring 2020	**MATH7604: Advanced Statistical Inference II**, A-
Fall 2019	**MATH7603: Advanced Statistical Inference I**, B-
Fall 2019	**PSY7001: Experimental Design**, A+
Fall 2019	**CSIE2120: Linear Algebra**, A+
Spring 2019	**PSY5033: Applied Linear Regression**, A+
Fall 2018	**EPM5074: Applied Bayesian Statistical Analysis**, A

Training

Statistical Teaching Assistant Training Course — *Taipei, Taiwan*

Statistics Education Center, National Taiwan University — *2020*

Functional Magnetic Resonance Imaging Training Course I — *Taipei, Taiwan*

Imaging Center for Integrated Body, Mind and Culture Research, College of Science, National Taiwan University — *January 2019*

Magnetoencephalography Training Course I — *Taipei, Taiwan*

Imaging Center for Integrated Body, Mind and Culture Research, College of Science, National Taiwan University — *January 2019*

Skills

Analytical Statistical modeling, Computational modeling, Bayesian statistics, Neuroimage analysis (fMRI and MEG)

Programming R (Tidyverse, R Markdown, and Shiny), JAGS, Stan, Python (PyTorch), MATLAB (SPM12 and Brainstorm), SAS, Linux

Tools Git, GitHub, LaTeX

Extracurricular Activities

The 9th Taiwan National Fire Dance Competition — *Taiwan (R.O.C.)*

Leader and Choreographer — *2017*

- Group champion

National Taiwan University Fire Dance Club — *National Taiwan University*

Vice President — *2016-2017*

The 11th Presentation of National Taiwan University Fire Dance Club — *National Taiwan University*

Coordinator and Choreographer — *2016*

The 10th Presentation of National Taiwan University Fire Dance Club — *National Taiwan University*

Vice Coordinator and Choreographer — *2016*

Forest Summer Camp — *Department of Forestry and Resources Conservation*

Lecturer and Team Leader — *2015, 2016*

Forest Orientation Camp — *Department of Forestry and Resources Conservation*

Vice Coordinator — *2015*

National Taiwan University Student Association — *National Taiwan University*

Member of the Election and Recalling Execution Commission — *2014-2015*